

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

Case Study

Code of Conduct:

I P.Sai Ganesh Reddy(192525111) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

P.Sai Ganesh Reddy

Annexure A

Short Answer Test

1, A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.

2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.

3. A telecommunications company aims to predict customer churn using historical usage data and service feedback. Using probabilistic models such as **Naïve Bayes**, **Bayesian inference**, and hypothesis space search, design a predictive system. Analyze how prior probabilities and feature dependencies influence churn prediction. Evaluate how uncertainty in customer behavior is handled and discuss the effectiveness of the model in improving customer retention strategies.

4. A smart healthcare system is designed to monitor patients in real-time using wearable sensors. Using **Bayesian Belief Networks** and probability learning, develop a model to detect anomalies in vital signs. Analyze how conditional dependencies between physiological parameters are modeled and how uncertainty due to sensor noise is handled. Evaluate system performance and discuss challenges in real-time deployment.

5. A logistics company wants to optimize delivery routes under uncertain traffic and weather conditions. Using **Bayesian inference** and probabilistic reasoning, design a system that predicts optimal routes dynamically. Analyze how prior knowledge and real-time data updates influence decision-making. Evaluate computational complexity and discuss how uncertainty impacts route optimization efficiency.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1, A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.

Intelligent Disease Diagnosis System

A hospital can develop a probabilistic diagnosis system using **Bayes' Theorem, Naïve Bayes, and Bayesian Belief Networks (BBN)**. The system takes symptoms, medical history, age, test results, and other patient information as input and estimates the probability of different diseases.

Bayes' Theorem can be used to calculate the probability of a disease given observed symptoms:

$$\mathbf{P(Disease \mid Symptoms) = [P(Symptoms \mid Disease) \times P(Disease)] / P(Symptoms)}$$

A Naïve Bayes classifier assumes that symptoms are conditionally independent given the disease. For example, fever, cough, and fatigue can be used as separate features to calculate the probability of influenza, pneumonia, or another disease.

A Bayesian Belief Network provides a more realistic model when symptoms are dependent. For example, **Disease → Fever → Heart Rate** can represent relationships between medical conditions and physiological symptoms.

Uncertainty is handled by assigning probabilities rather than making absolute decisions. Conditional dependencies improve prediction when symptoms are related. Probability learning allows the system to estimate probabilities from historical patient records.

The system can be evaluated using **accuracy, precision, recall, F1-score, and calibration**. In real-world healthcare, challenges include incomplete medical records, noisy measurements, privacy, changing disease patterns, and the need for reliable predictions.

Conclusion: A probabilistic model can support doctors by estimating disease probabilities while explicitly handling uncertainty and dependencies between symptoms.

2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.

Spam Detection System

An organization can develop an email spam detection system using **Naïve Bayes and the Bayes Optimal Classifier**. Email features such as word frequency, sender information, URLs, attachments, and message length can be extracted and used for classification.

Naïve Bayes calculates:

$P(\text{Spam} \mid \text{Features})$

and

$P(\text{Ham} \mid \text{Features})$

The email is classified into the class having the higher posterior probability.

The **Bayes Optimal Classifier** considers all possible hypotheses weighted according to their posterior probabilities. It can provide better theoretical performance when the probability distribution over hypotheses is accurately estimated, although it can be computationally expensive.

Maximum Likelihood Estimation (MLE) selects parameters that maximize the likelihood of the observed training data. In contrast, the **Minimum Description Length (MDL)** principle selects a model that provides a compact description of both the model and the data. MDL therefore helps control model complexity and reduce overfitting.

A larger sample size generally provides more reliable probability estimates. A very large hypothesis space increases model flexibility but can also increase overfitting and computational requirements.

Conclusion: Naïve Bayes provides a fast and effective spam classifier, while Bayes Optimal classification can provide theoretically strong predictions. MLE focuses on fitting observed data, whereas MDL balances data fitting with model simplicity.

3. A telecommunications company aims to predict customer churn using historical usage data and service feedback. Using probabilistic models such as **Naïve Bayes**, **Bayesian inference**, and hypothesis space search, design a predictive system. Analyze how prior probabilities and feature dependencies influence churn prediction. Evaluate how uncertainty in customer behavior is handled and discuss the effectiveness of the model in improving customer retention strategies.

Customer Churn Prediction

A telecommunications company can build a churn prediction system using **Naïve Bayes**, **Bayesian inference**, and **hypothesis-space search**. Historical features such as call duration, data usage, number of complaints, billing information, contract type, and customer feedback can be used.

Naïve Bayes calculates the probability of churn based on customer features:

$P(\text{Churn} \mid \text{Features})$

The prior probability of churn represents the overall likelihood that a customer will leave. If the historical churn rate is high, the prior probability will also be higher.

Feature dependencies can affect predictions. For example, **poor service quality** → **customer complaints** → **reduced satisfaction** → **churn** represents a dependency that may not be properly captured by the independence assumption of Naïve Bayes.

Bayesian inference updates prior beliefs when new customer information becomes available. Thus, a customer's churn probability can change when new complaints, usage patterns, or feedback are received.

Hypothesis-space search allows the organization to consider different predictive models and select a suitable hypothesis based on available evidence.

The system can be evaluated using **accuracy**, **precision**, **recall**, **F1-score**, and **ROC-AUC**. The predictions can help the company identify high-risk customers and provide targeted offers or improved services.

Conclusion: Probabilistic churn prediction helps the company handle uncertainty and identify customers who are likely to leave, supporting effective customer-retention strategies.

4. A smart healthcare system is designed to monitor patients in real-time using wearable sensors. Using **Bayesian Belief Networks** and probability learning, develop a model to detect anomalies in vital signs. Analyze how conditional dependencies between physiological parameters are modeled and how uncertainty due to sensor noise is handled. Evaluate system performance and discuss challenges in real-time deployment.

Smart Healthcare Monitoring System

A smart healthcare system can use wearable sensors to continuously monitor **heart rate, blood pressure, oxygen saturation, body temperature, and respiratory rate**.

A **Bayesian Belief Network** can represent dependencies among these physiological parameters. For example:

Health Condition → Heart Rate → Blood Pressure

and

Health Condition → Oxygen Saturation

The network assigns prior and conditional probabilities to the variables. When sensor readings are received, Bayesian inference updates the probability of an abnormal health condition.

Sensor noise and missing measurements introduce uncertainty. Instead of treating a sensor value as absolutely correct, the system can assign probabilities to possible states. Probability learning from historical patient data can improve these estimates.

The system can be evaluated using **accuracy, sensitivity, specificity, precision, recall, and false-alarm rate**. High sensitivity is particularly important because missing a serious abnormality can be dangerous.

Real-time deployment presents challenges such as limited computational resources, sensor failures, noisy data, communication delays, privacy, and the requirement for rapid responses.

Conclusion: A Bayesian Belief Network provides a suitable framework for real-time anomaly detection because it models dependencies between physiological parameters and handles uncertainty caused by sensor noise.

5. A logistics company wants to optimize delivery routes under uncertain traffic and weather conditions. Using **Bayesian inference** and probabilistic reasoning, design a system that predicts optimal routes dynamically. Analyze how prior knowledge and real-time data updates influence decision-making. Evaluate computational complexity and discuss how uncertainty impacts route optimization efficiency.

Dynamic Delivery Route Optimization

A logistics company can use **Bayesian inference and probabilistic reasoning** to dynamically select delivery routes under uncertain traffic and weather conditions.

The system can consider factors such as:

- Traffic congestion
- Weather conditions
- Road accidents
- Road closures
- Distance
- Delivery time
- Historical traffic patterns

Initially, the system uses **prior probabilities** based on historical traffic and weather information.

When real-time information becomes available, Bayesian inference updates these probabilities.

For example, if a particular route normally has a low probability of heavy traffic but a traffic sensor reports congestion, the posterior probability of delay increases. The system can then select an alternative route.

A probabilistic cost function can be used:

$$\text{Expected Cost} = \sum P(\text{Condition}) \times \text{Cost}(\text{Route}, \text{Condition})$$

The route with the lowest expected cost can be selected.

Real-time updates improve decision-making but increase computational requirements. A large number of routes and uncertain conditions can make exhaustive evaluation expensive. Efficient search and approximate inference methods can therefore be used.

Uncertainty can reduce route-optimization efficiency because predictions may change as new information arrives. However, continuously updating the probabilities allows the system to adapt to changing conditions.

Conclusion: Bayesian route optimization combines historical knowledge with real-time information to dynamically select efficient routes while accounting for uncertainty in traffic and weather.